

NBA Analytics Project — Formula & Calculation Reference

Final Reference, Per100 Edition — Seasons 2019-20 through 2024-25

Global Rules — Apply to All Calculations

- All composite scores use WEIGHTED ADDITION, not multiplication. Weights within each formula sum to 1.0.
- All rate-based inputs are computed on a Per100-possession basis throughout this project (never Per36).
- All raw metrics are converted to percentile ranks (0–100 scale) before use in composite scores. Exception: Points Created uses raw values.
- Percentile formula: $((\text{Total Players} - 1) / (\text{Rank} - 1)) \times 100$
- For lower-is-better stats (TOV%, DRtg, Opp FG%, Opp TS%, PF/100): Adjusted Percentile = $100 - \text{Percentile}$
- Minutes floor: 1,000 minutes played in a standard season, or 600 minutes in a season shortened by external circumstances. Below the floor, a player is excluded from award eligibility regardless of rate-stat performance.
- 65-Game minimum eligibility rule applies from 2023-24 onward, to every award and award team in this reference.
- Tiebreakers (in order): 1) Games Played 2) Minutes Played 3) Team Success Score 4) Higher seed

Part 1 — Foundation Scores (Individual Player)

These scores have no dependencies on other composite scores and are calculated first. All inputs are percentile-ranked before use unless noted otherwise.

1.1 Points Created [Volume — Raw Values, No Percentile]

Points Created	$\text{PTS} + (\text{AST} \times 2.2)$
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Note: The only formula using raw stats. The 2.2 multiplier estimates points generated per assist. Used as a volume signal in several award formulas.

1.2 Usage-Adjusted Efficiency (UAE)

UAE	$\text{TS\% percentile} \times (1 + (\text{USG\% percentile} / 100) \times 0.15)$
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Note: Rewards players who maintain efficiency at high usage. The 0.15 cap prevents usage from overwhelming the efficiency signal.

1.3 Scoring Efficiency Score

Scoring Eff.	$(0.50 \times \text{TS\%}) + (0.25 \times \text{FT Rate}) + (0.15 \times \text{PPSA}) + (0.10 \times \text{3PA Rate})$
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Note: eFG% is not included — it is already captured inside TS%. Weight redistributed to TS%.

1.4 Playmaking Score

Playmaking	$(0.40 \times \text{AST\%}) + (0.25 \times \text{AST/TOV}) + (0.20 \times \text{AST/100}) + (0.15 \times \text{ORTg})$
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Note: USG% is not included — it measures ball dominance, not playmaking quality. ORtg is retained because elite playmakers elevate team offense.

1.5 Defensive Impact Score

Def. Impact	$(0.40 \times \text{DBPM}) + (0.30 \times \text{DWS}) + (0.15 \times \text{STL}\%) + (0.15 \times \text{BLK}\%)$
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Note: Team DRtg context is intentionally excluded here — it appears at the award level instead, so good defenders on bad defensive teams aren't double-penalized.

1.6 Defensive Activity Score

Def. Activity	$(0.35 \times \text{STL}\%) + (0.25 \times \text{STOCKS}/100) + (0.20 \times \text{BLK}\%) + (0.15 \times \text{DREB}/100) + (0.05 \times \text{PF}/100 \text{ inverted})$
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Note: PF/100 is inverted — fewer fouls per 100 possessions scores higher. Weight kept small (0.05) because fouling is partly a style and role artifact.

1.7 Rim Protection Score

Rim Prot.	$(0.55 \times \text{BLK}\%) + (0.45 \times \text{Opp 0–3 FG}\% \text{ inverted})$
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Note: Opp 0–3 FG% sourced from the opponent-shooting table. Inverted — lower opponent FG% at the rim means better rim protection.

1.8 Perimeter Defense Score

Perim. Def.	$(0.70 \times \text{STL}\%) + (0.30 \times \text{STL}/100)$
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Note: Deflections are not available in the source data — weight redistributed fully to steal metrics.

Part 2 — Second-Level Scores (Individual Player)

These scores depend on the Part 1 foundation scores and are built after those views exist.

2.1 Offensive Impact Score

Off. Impact	$(0.30 \times \text{OBPM}) + (0.25 \times \text{ORtg}) + (0.20 \times \text{OWS}) + (0.15 \times \text{UAE}) + (0.10 \times \text{Points Created percentile})$
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2.2 Player Value Score

Player Value	$(0.30 \times \text{Offensive Impact Score}) + (0.25 \times \text{Defensive Impact Score}) + (0.20 \times \text{VORP}) + (0.15 \times \text{WS}/48) + (0.10 \times \text{Net Rating})$
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Note: Rebuilt from an earlier multiplicative version that produced near-zero outputs for nearly every player. VORP and WS/48 anchor this score as the most reliable all-in-one value metrics available.

Part 3 — Team Scores

All team metric inputs are percentile-ranked within each season before use in these formulas.

3.1 Team Offense Score

Team Off.	$(0.35 \times \text{ORTg}) + (0.25 \times \text{TS\%}) + (0.20 \times \text{eFG\%}) + (0.10 \times \text{AST\%}) + (0.10 \times \text{3PA Rate})$
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3.2 Team Defense Score

Team Def.	$(0.35 \times \text{DRtg inverted}) + (0.25 \times \text{Opp eFG\% inverted}) + (0.15 \times \text{Opp TS\% inverted}) + (0.15 \times \text{DREB\%}) + (0.10 \times \text{Opp TOV\%})$
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Note: Opp TOV% is NOT inverted — higher opponent turnovers means better defense.

3.3 Team Success Score

Team Success	$(0.40 \times \text{Net Rating}) + (0.30 \times \text{Win\%}) + (0.15 \times \text{Team Offense Score}) + (0.15 \times \text{Team Defense Score})$
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Note: Win% = wins / (wins + losses), calculated inline from the team advanced-stats fact table.

Part 4 — Individual Award Formulas

4.1 MVP (Team Influence)

MVP	$(0.25 \times \text{Player Value Score}) + (0.20 \times \text{Offensive Impact Score}) + (0.15 \times \text{Defensive Impact Score}) + (0.15 \times \text{Team Success Score}) + (0.15 \times \text{Scoring Eff. Score}) + (0.10 \times \text{Playmaking Score})$
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Note: Team Success Score carries meaningful weight (0.15) because real MVP voting heavily rewards players on winning teams. The no-team-influence variant was cut — see the Final Award Set document.

4.2 DPOY — Defensive Player of the Year

DPOY	$(0.35 \times \text{Defensive Impact Score}) + (0.25 \times \text{Team Defense Score}) + (0.20 \times \text{Defensive Activity Score}) + (0.10 \times \text{Rim Protection Score}) + (0.10 \times \text{Perimeter Defense Score})$
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Note: Team Defense Score gets 0.25 by design — DPOY voters heavily reward players whose teams rank elite defensively.

4.3 OPOY — Offensive Player of the Year

OPOY	$(0.40 \times \text{Offensive Impact Score percentile}) + (0.25 \times \text{Scoring Eff. Score}) + (0.20 \times \text{Playmaking Score}) + (0.15 \times \text{Points Created percentile})$
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Note: Unlike DPOY, OPOY carries no team-context weighting. This is intentional — OPOY is scored as a pure individual-skill award, consistent with the fictional awards in Part 5, rather than mirroring DPOY's deliberate reward for players on elite defensive teams. Confirmed directly from the live SQL view definition.

Part 5 — Fictional Individual Awards

Fictional awards are independent of team record. Team Success Score does not appear in any formula below.

5.1 Best Player

Best Player

$$(0.40 \times \text{Player Value Score}) + (0.30 \times \text{Offensive Impact Score}) + (0.20 \times \text{Defensive Impact Score}) + (0.10 \times \text{BPM})$$

5.2 Best Scorer [Efficiency Factored In]

Best Scorer

$$(0.45 \times \text{Scoring Eff. Score}) + (0.30 \times \text{PTS/100 percentile}) + (0.15 \times \text{Usage-Adjusted Efficiency}) + (0.10 \times \text{TS\%})$$

5.3 Best Playmaker

Best Playmaker

$$(0.45 \times \text{Playmaking Score}) + (0.25 \times \text{AST\%}) + (0.20 \times \text{AST/TOV}) + (0.10 \times \text{ORTg})$$

Part 6 — Award Team Formula

6.1 All-NBA Score [Used for both Positioned and Positionless]

All-NBA

$$(0.30 \times \text{Offensive Impact Score}) + (0.25 \times \text{Player Value Score}) + (0.20 \times \text{Defensive Impact Score}) + (0.15 \times \text{Points Created percentile}) + (0.10 \times \text{Scoring Eff. Score})$$

All-NBA Team Selection:

- Positioned (pre-2023-24): 1st Team — #1 & #2 Guards, #1 & #2 Forwards, #1 Center | 2nd Team — #3–4 Guards, #3–4 Forwards, #2 Center | 3rd Team — #5–6 Guards, #5–6 Forwards, #3 Center
- Positionless (2023-24 onward): 1st Team — Top 5 players | 2nd Team — #6–10 | 3rd Team — #11–15
- 65-game minimum applies from 2023-24 onward.

Note: All-Defense Teams and All-Rookie Teams used closely related formulas during development but were cut from the final award set — see the Final Award Set document for the reasoning.

Part 7 — Fictional Team Awards

7.1 Best Offensive Team

Best Off. Team

Team Offense Score [direct score comparison, no additional formula]

7.2 Best Defensive Team

Best Def. Team

Team Defense Score [direct score comparison, no additional formula]

7.3 Most Efficient Team

Most Efficient

$(0.45 \times \text{TS\%}) + (0.30 \times \text{eFG\%}) + (0.15 \times \text{TOV\% inverted}) + (0.10 \times \text{FT Rate})$

Note: TOV% inverted — a lower turnover rate produces a higher efficiency score.

7.4 Best Overall Team

Best Overall

Team Success Score [direct score comparison, no additional formula]

Part 8 — Percentile Engine Reference

Every scored metric below is converted to a 0–100 percentile rank within each season before being used in any composite score.

Standard Percentile Formula

Percentile

$((\text{Total Players} - 1) / (\text{Rank} - 1)) \times 100$

Note: Rank 1 = best player. Applied per season using a PostgreSQL window function.

Inverted Percentile — Lower Is Better

Inverted

$100 - \text{Percentile}$

Note: Applies to: TOV%, DRtg, Opp FG%, Opp TS%, Opp eFG%, PF/100, Opp 0–3 FG%.

Metrics Requiring Percentile Conversion Before Use

- OFFENSIVE: TS%, FT Rate, PPSA, 3PA Rate, AST%, AST/TOV, AST/100, ORtg, OBPM, OWS, VORP, WS/48, Net Rating, PTS/100, BPM, UAE (itself a composite)
- DEFENSIVE: DBPM, DWS, STL%, BLK%, STOCKS/100, DREB/100, PF/100 (inverted)
- TEAM: ORtg, TS%, eFG%, AST%, 3PA Rate, DRtg (inverted), Opp eFG% (inverted), Opp TS% (inverted), DREB%, Opp TOV%, Net Rating, Win%
- NOT percentiled: Points Created (raw values only)

Part 9 — Build Order & Dependencies

Views are built in this order. Each layer depends only on the layer(s) before it.

Step	View / Score	Dependencies
1	v_percentiles_player	Raw fact tables (all player fact tables)
2	v_percentiles_team	Raw fact tables (all team fact tables)
3	Points Created	fact_player_season_totals (raw PTS, AST)
4	UAE	v_percentiles_player (TS%, USG%)
5	Scoring Efficiency Score	v_percentiles_player
6	Playmaking Score	v_percentiles_player
7	Defensive Impact Score	v_percentiles_player
8	Defensive Activity Score	v_percentiles_player
9	Rim Protection Score	v_percentiles_player + v_percentiles_team (Opp 0–3 FG%)
10	Perimeter Defense Score	v_percentiles_player
11	Team Offense Score	v_percentiles_team
12	Team Defense Score	v_percentiles_team
13	Team Success Score	Team Offense Score + Team Defense Score + v_percentiles_team
14	Offensive Impact Score	Steps 4 + 3 (UAE + Points Created)
15	Player Value Score	Steps 14 + 7 (Off. Impact + Def. Impact) + v_percentiles_player
16	OPOY	Steps 14, 5, 6, 3 (Off. Impact percentile + Scoring Eff. + Playmaking + Points Created percentile)
17	All remaining award formulas	All steps above complete